Data Preprocessing, Quality Control, and Normalization Strategies for EV Uptake Proteomics

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1. Introduction and Experimental Framework

Biological Context and Research Problem

Metastatic dissemination of prostate cancer exhibits an overwhelming tropism for the skeleton, with bone metastases developing in the vast majority of patients with advanced disease. Before macroscopic colonization occurs, primary tumors remodel distant microenvironments by establishing pre-metastatic niches. Extracellular vesicles (EVs) act as critical mediators of this systemic communication.

While it is established that tumor-derived EVs condition stromal populations, a fundamental mechanistic question remains unresolved: is EV uptake by recipient bone cells driven by non-specific, stochastic bulk endocytosis, or do metastatic vesicles possess specialized surface and cargo programs that confer selective tropism and high-affinity internalization?

Experimental System and Objectives

To delineate the molecular drivers governing vesicle internalisation, we established a quantitative proteomic model comparing parental PC3 and highly metastatic human prostate cancer cell lines:

- **Donor Vesicles:** Donor Vesicles: EVs harvested from parental PC3 cells and their highly bone-metastatic subline PC3M (PC3M-luc-C6).
- **Recipient Microenvironment:** Human osteoblast-like osteosarcoma cells (SAOS-2), providing a cellular model of the receptive bone niche.

The primary objective of this study is to implement a rigorous computational proteomics pipeline capable of isolating authentic vesicular cargo from cellular background, resolving severe loading imbalances, and identifying specific protein signatures that distinguish metastatic uptake programs.

Multiplexed Quantitative Strategy (SILAC-TMT-RTS-SPS-MS3)

Dissecting internalized low-abundance EV cargo from the massive endogenous proteome of recipient osteoblasts presents an extreme analytical challenge. To achieve absolute quantification and prevent cross-contamination, we employed a hybrid isotopic labeling framework combining metabolic pulse-labeling (SILAC) with high-multiplex isobaric tagging (TMTpro 18-plex):

1. **Metabolic Discrimination:** Donor prostate cancer cells were cultured under Heavy isotope conditions (Heavy Arg10/Lys-D8). All genuine EV cargo retains a diagnostic mass shift, whereas endogenous recipient osteoblast proteins remain Light.
2. **Isobaric Multiplexing:** Samples were multiplexed into a single TMTpro 18-plex run, eliminating inter-run chromatographic drift and instrument variability.
3. **Targeted Acquisition:** Real-time search (RTS) duty-cycle filtering coupled with synchronous precursor selection (SPS-MS3) targeted exclusively Heavy precursor peptides, lowering reporter ion ratio compression from co-isolated Light background.

The functional layout of the 18 TMT channels is structured as shown in **Table 1**:

Table 1. Experimental layout and functional allocation of the TMTpro 18-plex run.

Function	Channels	Sample / Content
Natural Uptake	130N, 130C, 131N	SAOS-2 + PC3 EV
	131C, 132N, 132C, 135N	SAOS-2 + PC3M EV
Spike-in Reference	133N, 133C	SAOS-2 lysate + PC3 EV
	134N, 134C	SAOS-2 lysate + PC3M EV
Negative Controls	128C, 129N, 129C	SAOS-2 alone (background)
	127C, 128N	Empty
Boost Channels*	126, 127N	Pure concentrated EVs

**Boost channels provided ion current to drive RTS acquisition and were omitted from all comparative uptake metrics.*

2. Data Curation and Peptide Preprocessing Workflow

Quantitative evaluation was conducted on the PSM dataset (comprising all analytical fractions, n = 88,138 peptide-spectrum matches; dataset file: *MEK_25_02_label_included.txt*), spanning the complete set of multiplexed TMT channels, biological controls, and spike-in reference standards. Individual fraction files and alternative acquisition parameter sets were excluded to ensure a uniform data processing workflow focused on biological uptake differences.

Identification Confidence and Quality Control

Prior to quantitative aggregation, dataset integrity was evaluated across four primary quality control metrics: metabolic incorporation, contaminant detection, mapping uniqueness, and identification false discovery rate (FDR) (Table 2, Figure 1).

All 88,138 records passed search-engine filtering with $q \leq 0.01$ (FDR 1%; Figure 1D), with zero entries flagged as common laboratory contaminants (Figure 1B). Light peptides from the recipient SAOS-2 cellular background accounted for 97.5% of entries (n = 85,929), while heavy-labeled peptides from vesicular cargo accounted for 2.5% (n = 2,209) (Figure 1A). Furthermore, 97.7% of the sequences mapped uniquely to a single master protein accession (Figure 1C).

Table 2. Baseline Filtering Summary. *Delineates the input dataset structure at the peptide level prior to abundance filtering.*

QC Metric	Value / Criterion	Peptide Count (n)	Share of Pool (%)
Total Identifications	Raw output	88,138	100.0%
PSM Spectral Confidence	q-value ≤ 0.01 (FDR 1%)	88,138	100.0%
Contaminants	Excluded	88,138	100.0%
Mapping Specificity	Unique (1 protein)	86,106	97.7%
	Shared / Multimapping (> 1)	2,032	2.3%
Metabolic Origin	Light (recipient cells)	85,929	97.5%
	Heavy (EV cargo)	2,209	2.5%

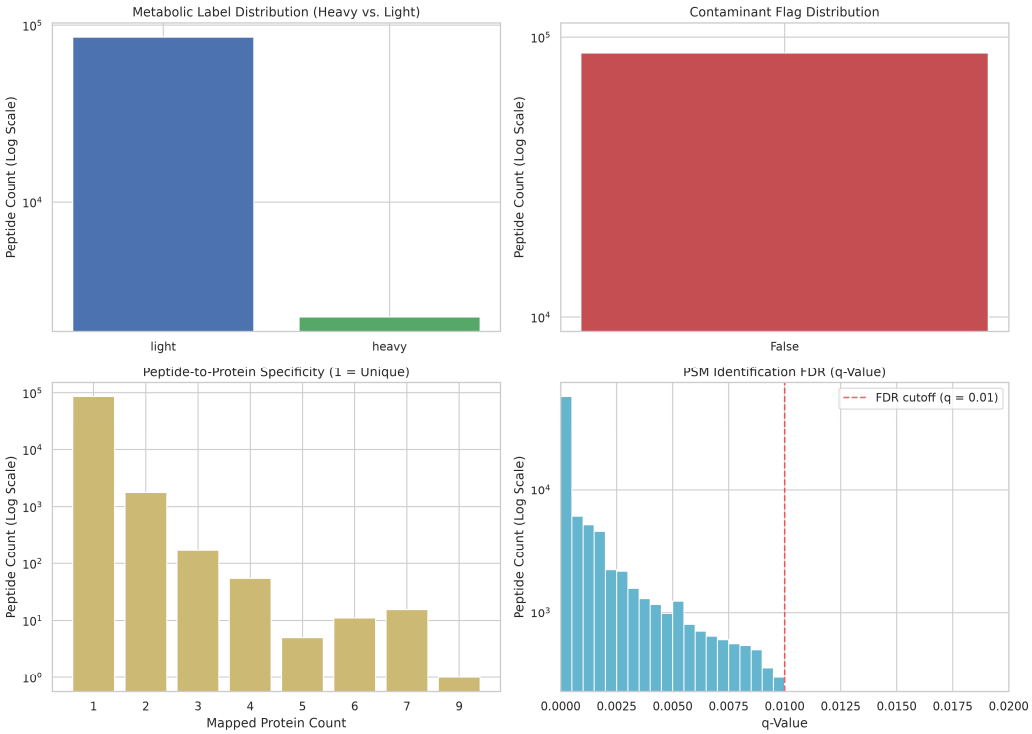


Figure 1. Initial spectral library curation, identification quality control, and peptide specificity.
(A) *Distribution of metabolic labels across all identified peptide-spectrum matches (PSMs; total N = 88,138). Endogenous recipient background (Light) comprises 97.5% (n = 85,929), while Heavy-labeled EV cargo constitutes 2.5% (n = 2,209).*
(B) *Contaminant status for each peptide; 100% of identifications (n = 88,138) represent authentic target sequences.*
(C) *Sequence mapping specificity: 86,106 peptides (97.7%) mapped uniquely to a single master protein accession, while 2,032 peptides mapped to multiple proteins.*
(D) *Identification confidence: 100% of accepted entries fulfilled $q \leq 0.01$ (median $q = 1.05 \times 10^{-4}$).*

Proteolytic Digestion Fidelity

Enzymatic cleavage reproducibility was assessed by evaluating sequence length profiles across the identified peptide populations (Figure 2). Peptide lengths ranged from 7 to 30 amino acids, with a median of 12 amino acids and a mean of 13.13 amino acids (Figure 2A). Kernel density estimation demonstrated near-identical length distributions between Light cellular background (n = 85,929) and heavy vesicular cargo (n = 2,209) showed matching distributions (Figure 2B), confirming consistent tryptic digestion across both pools without size-dependent recovery artifacts.

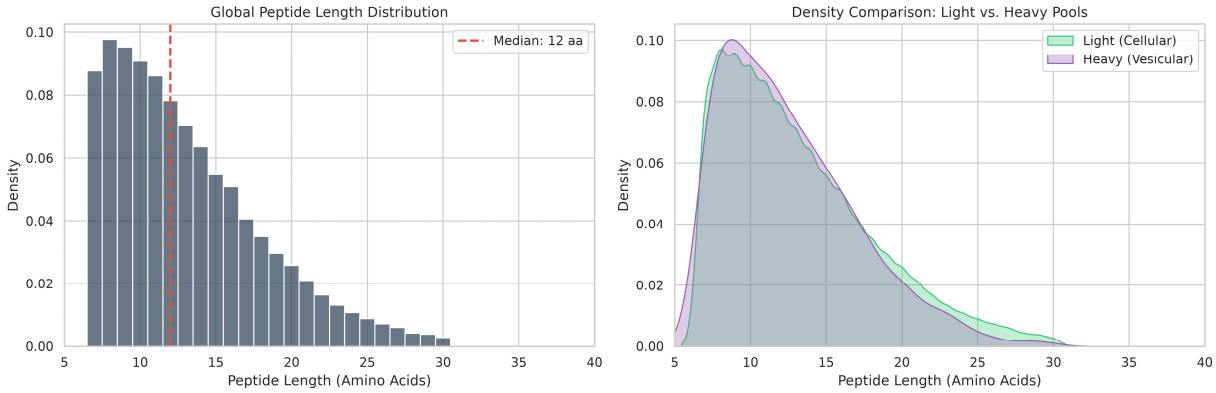


Figure 2. Peptide sequence length distribution across identified pools.
(A) *Global length distribution for all identified peptides (N = 88,138; median = 12 amino acids; mean = 13.13 amino acids).*
(B) *Kernel density estimation of peptide lengths for Light (endogenous cellular, n = 85,929) vs. Heavy (vesicular cargo, n = 2,209) peptide pools, indicating uniform enzymatic digestion across both compartments.*

Reporter Channel Occupancy and RTS Duty-Cycle Constraints

Channel occupancy was evaluated across the 18 multiplexed TMT channels (Figure 3). Because real-time search (RTS) triggering specifically targeted Heavy precursor ions, 97.7% of Light background peptides (n = 83,963) exhibited zero active reporter channels. The remaining 2.3% of Light peptides (n = 1,966) displayed nominal signal arose from precursor co-isolation within the quadrupole isolation window during multiplexed acquisition.

Among the 2,209 Heavy vesicular peptides, 1,018 features yielded quantifiable reporter signal, forming an occupancy distribution centered around 14 active channels (Figure 3).

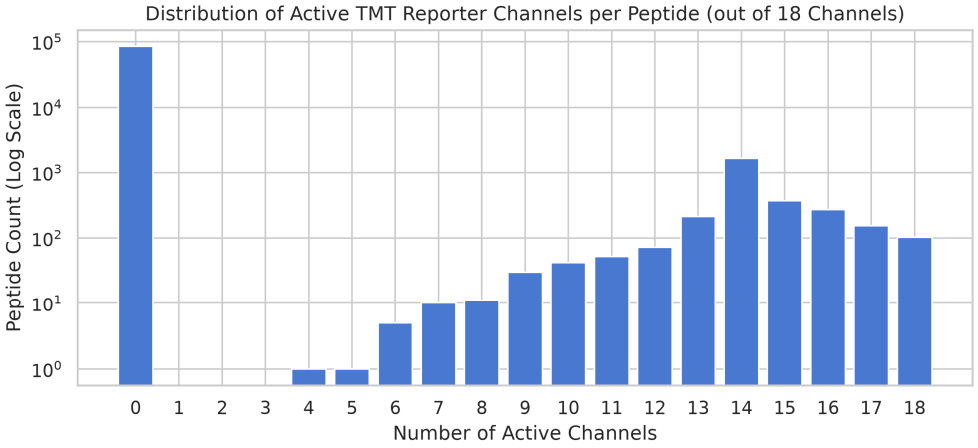


Figure 3. TMT reporter channel occupancy per peptide across the multiplex. *Distribution of active reporter channels (out of 18 total multiplexed channels) across all identified peptides (N = 88,138). Unquantified Light background dominates the zero-occupancy bin (n = 85,551, comprising 83,963 Light background peptides and 1,191 unquantified Heavy precursors; mean active channels = 0.32). Quantifiable features display a bell-shaped distribution centered near 14 channels, driven by the Heavy vesicular cargo pool (mean active channels = 6.63, n = 2,209).*

To investigate the origin of unquantified Heavy peptides, instrument acquisition status flags (Quan Info) were systematically profiled (**Table 3**):

Table 3. Investigation of Unquantified Heavy Peptides (Quan Info).

Status Flag	Peptide Count (n)	Share of Heavy (%)	Analytical Cause
Passed (No Flags)	1,018	46.1%	Valid SPS-MS3 acquisition satisfying all quantitative criteria.
NoQuanValues	1,175	53.2%	Confirmed at MS2, but MS3 was skipped due to RTS duty-cycle saturation.
ExcludedByMethod	16	0.7%	Acquired at MS3 but rejected due to high co-isolation interference or low SPS match.

The primary driver of missing quantitative values was chromatographic duty-cycle saturation under RTS operation: 53.2% of Heavy peptides (n = 1,175) were identified by MS2 spectra but did not trigger an SPS-MS3 scan. Only 0.7% (n = 16) were discarded due to co-isolation thresholds. Thus, 1,018 Heavy peptides provided high-fidelity quantitative reporter signals.

Carrier Artifact Depletion and Protein Rollup Specificity

To prevent carrier-driven false discoveries, the 1,018 quantifiable Heavy peptides were screened across all functional uptake co-culture channels (Channels 130N–132C, 135N). Exactly 7 peptides showed reporter intensity restricted exclusively to EV boost channels (Channels 126, 127N) or spike-in reference mixtures (Channels 133N–134C), with zero signal in recipient cell uptake channels. Excluding these 7 non-informative features yielded a core dataset of 1,011 Heavy cargo peptides.

Within the Heavy peptide pool, 73 peptides mapped to multiple related protein accessions. Detailed manual curation classified these shared features into three distinct biological and genetic classes:

- **Alternative splicing isoforms** of identical gene loci (e.g., PKM, HNRNPC).
- **Gene duplication events or pseudogenes** sharing identical mature sequences (e.g., ANXA2, and core histones including H2AC11 and H3-3A).
- **Conserved paralogous gene families** (e.g., ATP1A1, ACTB/ACTC1).

While collapsing multi-mapping peptides into grouped gene-level clusters (e.g., *ACTB/ACTC1* or *PKM/PKM2*) represents a viable strategy to salvage ~20 biologically relevant proteins without loss of functional interpretation, we elected to apply a strictly conservative baseline for this primary report. All 73 multi-mapping peptides were set aside, restricting quantitative protein-level rollup (computed by summing unique peptide intensities per master accession) strictly to peptides mapping uniquely to a single master accession.

This stringent rollup protocol established an initial baseline cohort of 372 unambiguously quantified EV cargo proteins, ensuring that downstream statistical modeling is free from isoform assignment bias. Gene-level group collapsing for the remaining 20 multi-mapping proteins remains designated for subsequent pipeline refinement.

3. Background Gating, Noise Floor Filtering, and Technical Outlier Curation

To distinguish genuine EV cargo proteins from non-specific background signals, instrument electronic noise, and endogenous recipient cell proteome, an empirical dual-threshold filtering strategy was implemented.

In multiplexed SPS-MS3 isobaric labeling workflows, non-zero reporter ion signals can emerge in negative control channels due to isotopic impurities, mild quadrupole co-isolation cross-talk, and baseline detector noise. Furthermore, raw intensity distributions exhibit pronounced right-skewness and left-censoring near the instrument's physical limit of detection (LOD ≈ 1.7, defined as the minimum non-zero reporter intensity detected across any multiplex channel), rendering standard parametric normal models inapplicable. Accordingly, robust non-parametric metrics were utilized. The filtering framework relies on two complementary criteria:

1. **Absolute Noise Floor Threshold (α):**

The baseline noise profile was empirically estimated using non-zero reporter intensities across negative biological controls (recipient SAOS-2 cells alone, n = 3) and empty instrument buffer channels (n = 2). Across these background channels, the median signal was 6.55 in SAOS-2 controls and 7.00 in empty channels (with 75th percentiles of 12.90 and 14.90, respectively). Given the matching intensity distributions between empty instrument channels and non-treated recipient cell controls, this low-level signal represents non-specific analytical background. Consequently, the baseline noise floor was set to the empirical background median (α = 6.6). A protein was required to exceed this noise floor in at least one experimental condition:

$$\max(\text{Median}_{\text{PC3}}, \text{Median}_{\text{PC3M}}) > \alpha$$

To rigorously validate α and evaluate potential trade-offs between background exclusion and genuine cargo preservation, empirical kernel density estimations (KDE) and cumulative exclusion profiles were computed (Figure 4). The intensity distributions of EV uptake channels (PC3 median = 11.0, PC3M median = 13.0) and negative controls (SAOS-2 median = 6.55, empty median = 7.00) overlap smoothly in the low-intensity regime without displaying bimodal separation or a distinct local minimum between signal and noise (Figure 4A). An empirical density intersection between background and uptake occurs at α_{valley} = 12.63.

Crucially, setting an exclusion threshold at this density intersection (α_{valley} = 12.63) would result in severe false-negative depletion, discarding 54.81% and 49.16% of individual intensity measurements in PC3 and PC3M channels, respectively (Figure 4B). At the protein cohort level, enforcing α_{valley} would discard 181 out of 372 quantified proteins (48.66%), eliminating nearly half of the proteome before differential hypothesis testing. Furthermore, cumulative exclusion curves reveal that biological background (SAOS-2 alone) and uptake channels scale quasi-parallelly with respect to threshold tightening, confirming the absence of an objective inflection point. Selecting the baseline noise median (α= 6.6) provides an optimal, high-sensitivity pre-filter: it eliminates 50.72% of recipient background observations and 23.66% of low-intensity protein candidates while preserving >72% of authentic EV uptake measurements (Figure 4B). Complete discrimination between genuine biological cargo and endogenous recipient

background is subsequently resolved by the secondary fold-change criterion (β), preventing catastrophic feature loss during initial gating.

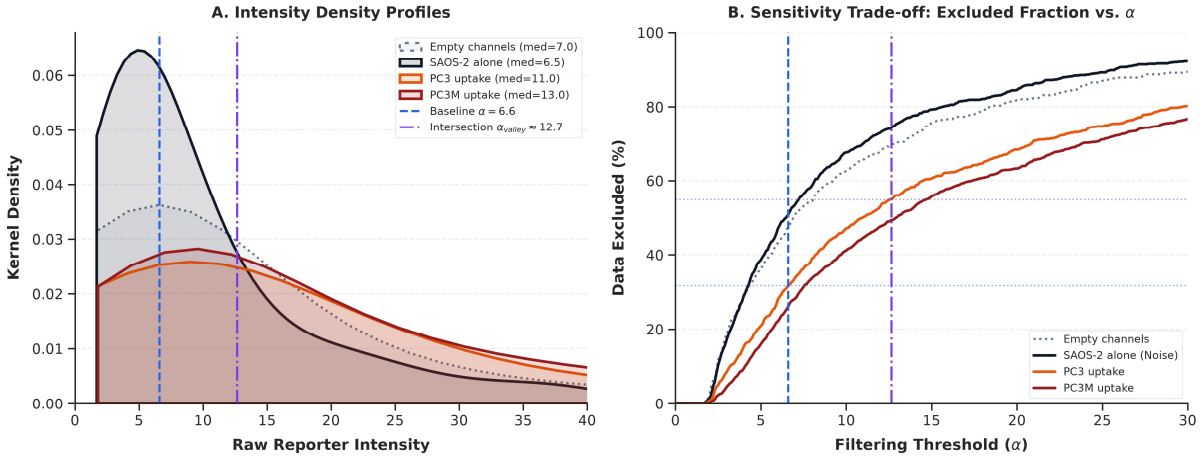


Figure 4. Reporter ion intensity distribution profiles and sensitivity trade-off analysis.

(A) *Kernel density estimates of raw reporter ion intensities across empty instrument channels (n = 2, dotted grey), biological recipient background (SAOS-2 alone, n = 3, dark navy), and EV uptake channels (PC3, n = 2, orange; PC3M, n = 4, dark red). Vertical dashed lines denote the adopted baseline noise floor (α = 6.60, blue dashed) and the empirical density intersection point (α_{valley} = 12.63, purple dash-dotted).*

(B) *Empirical cumulative exclusion curves demonstrating data attrition as a function of threshold parameter α. Applying α_{valley}= 12.63 eliminates 74.22% of background but inappropriately purges 51.01% of combined uptake observations (54.81% in PC3 and 49.16% in PC3M; 48.66% of candidate proteins). The adopted noise floor (α = 6.60) filters 50.72% of background noise while preserving 72.17% of all EV uptake observations (retaining 284 proteins, 75.5%) for downstream fold-change evaluation.*

2. **Signal-to-Background Ratio (β):**

To protect against analytical fluctuations and technical variability, the fold-change threshold (β) was derived directly from experimental reproducibility. The median coefficient of variation (CV = SD/Mean) across recipient control replicates was determined to be 24%. Under standard error propagation, the relative technical uncertainty of the ratio between two independent noisy measurements (sample vs. control) is:

$$CV_{\text{ratio}} \approx \sqrt{CV_{\text{sample}}^2 + CV_{\text{control}}^2} = \sqrt{0.24^2 + 0.24^2} \approx 0.34\ (34\%)$$

To achieve statistical confidence at the 3σ level ($p < 0.003$), an authentic cargo signal must exceed random analytical fluctuations by at least three standard deviations of the ratio ($3 \times 34\% \approx 102\%$, representing a >2-fold increase). Thus, the fold-change threshold was established at $\beta = 2.0$ ($\log_2 FC \geq 1.0$):

$$\max(\text{Median}_{\text{PC3}}, \text{Median}_{\text{PC3M}}) \geq \beta \times \text{Median}_{\text{SAOS2 control}}$$

From an initial pool of 372 strictly quantified proteins, 284 proteins (76.3%) satisfied the absolute baseline noise threshold ($\alpha = 6.6$), filtering out 88 near-LOD artifacts. Subsequently, enforcing the dual criteria (α = 6.6 and β = 2.0) retained a cohort of 212 validated EV cargo proteins (57.0% of the initial pool), discarding 72 features as non-specific background or ambiguous signals within experimental error margins.

Technical Outlier Identification and Sample Curation

Prior to comparative analysis, channel-wise metrics were evaluated across all multiplexed co-culture channels. Channel *PC3_EV_R1* exhibited severe analytical suppression indicative of library preparation failure: cumulative reporter intensity was reduced by more than 3.5-fold compared to parallel biological replicates (2,266.9 vs. 7,448.2 and 8,614.1 in *PC3_EV_R2* and *PC3_EV_R3*, respectively), channel median intensity collapsed to baseline detector noise (2.90), and zero-value dropouts surged to 33.1% (compared to ≤ 7.0% in intact replicates).

Crucially, global total signal normalization cannot recover *PC3_EV_R1*: scaling up an underlabeled channel with one-third missing values would simply amplify electronic noise and impute artificial high-variance noise into downstream statistics. Consequently, *PC3_EV_R1* was classified as a technical failure and excluded from downstream analysis. Subsequent processing of the 212 authenticated cargo proteins was performed across validated replicates (n=2 for PC3-EV, n=4 for PC3M-EV).

4. Internal Spike-in Normalization and Relative Cargo Selectivity

Raw reporter ion intensities in shotgun mass spectrometry reflect a convolution of absolute molecular abundance and sequence-specific physicochemical properties, such as tryptic cleavage kinetics, electrospray ionization efficiency, and parent vesicle abundance. Consequently, a high raw intensity in an uptake channel can represent either efficient cellular internalization of a low-abundance cargo or, conversely, inefficient, passive uptake of an abundant, highly ionizable protein.

Condition-Matched Spike-in Normalization

To decouple genuine internalization efficiency from mass spectrometric responsiveness, uptake intensities were normalized against matched spike-in reference channels. These spike-in channels consist of an unincubated mixture of recipient cell lysate and authentic donor EVs, capturing baseline abundance, peptide yield, and ionization characteristics of each protein species prior to biological co-culture.

For each condition, technical spike-in replicates (n = 2; channels 133N/C for PC3, and 134N/C for PC3M) were averaged to establish a stable reference baseline:

$$\bar{I}_{\text{Spike-in},j} = \frac{I_{Spike-in,1j} + I_{Spike-in,2j}}{2}$$

The relative uptake metric for protein i in experimental channel j is calculated as the ratio of sample intensity to this matched spike-in reference:

$$\text{Relative Uptake}_{ij} = \frac{I_{ij}}{\bar{I}_{\text{Spike-in},j}}$$

This ratio transforms raw ion intensities into a dimensionless relative uptake metric, decoupling biological uptake from baseline vesicle abundance and individual peptide ionization efficiency. Anchoring each protein to its condition-matched input standard enables direct comparison of internalization efficiency across different protein species within the vesicle payload.

Mathematical Invariance to Particle Dosing Offsets

Nanoparticle Tracking Analysis (NTA) indicated an approximate $2\times$ higher particle counts for PC3 EV suspensions relative to PC3M preparations. While a nominal scaling offset ($\Delta\log_2 = -1.0$) can be introduced to reflect physical particle counts, empirical enumeration methods carry substantial heuristic variance.

Crucially, given the nature of our downstream analytical pipeline, where sample-median centering is applied prior to ordination and statistical testing, the outcome is mathematically independent of condition-specific scalar offsets. For any uniform additive constant C in $\log_2$-space, centering yields:

$$(y_{ij} - C) - \text{median}_j(y_j - C) = y_{ij} - \text{median}_j(y_j)$$

Because any uniform dosing correction cancels out algebraically, downstream evaluations of relative cargo selectivity and differential uptake remain completely invariant to heuristic particle enumeration uncertainties.

Transformation and Data Completeness

Logarithmic transformation was applied strictly to genuine non-zero abundance measurements

$$y_{ij} = \log_2\left(\text{Relative Uptake}_{ij}\right)$$

without using global pseudocounts (such as $\log_2(I + 1)$), which introduce non-linear compression and distort true fold changes in low-intensity regimes.

Across the validated cohort (N = 212 proteins across 6 uptake channels; 1,272 data points), data completeness was exceptionally high at 99.85%. Only 2 isolated values (0.15%) dropped below detection thresholds and were conservatively substituted with the limit of detection halved (LOD/2 ≈ 0.85), preserving ratio linearity without distorting sample covariance.

5. Exploratory Ordination and Resolution of Global Bulk Uptake Effects

Unsupervised Principal Component Analysis (PCA) and pairwise Pearson correlation analyses were performed on the spike-normalized, $\log_2$-transformed intensity matrix across the authenticated cargo cohort (N = 212 proteins, 6 biological channels).

High technical and biological concordance was confirmed within each experimental condition. Parental PC3 replicates displayed strong inter-replicate correlation (r = 0.917), while metastatic PC3M replicates formed an internally consistent cohort (pairwise r = 0.76 - 0.94) that distinctly segregated from the PC3 condition (r = 0.33 - 0.57) (Figure 5).

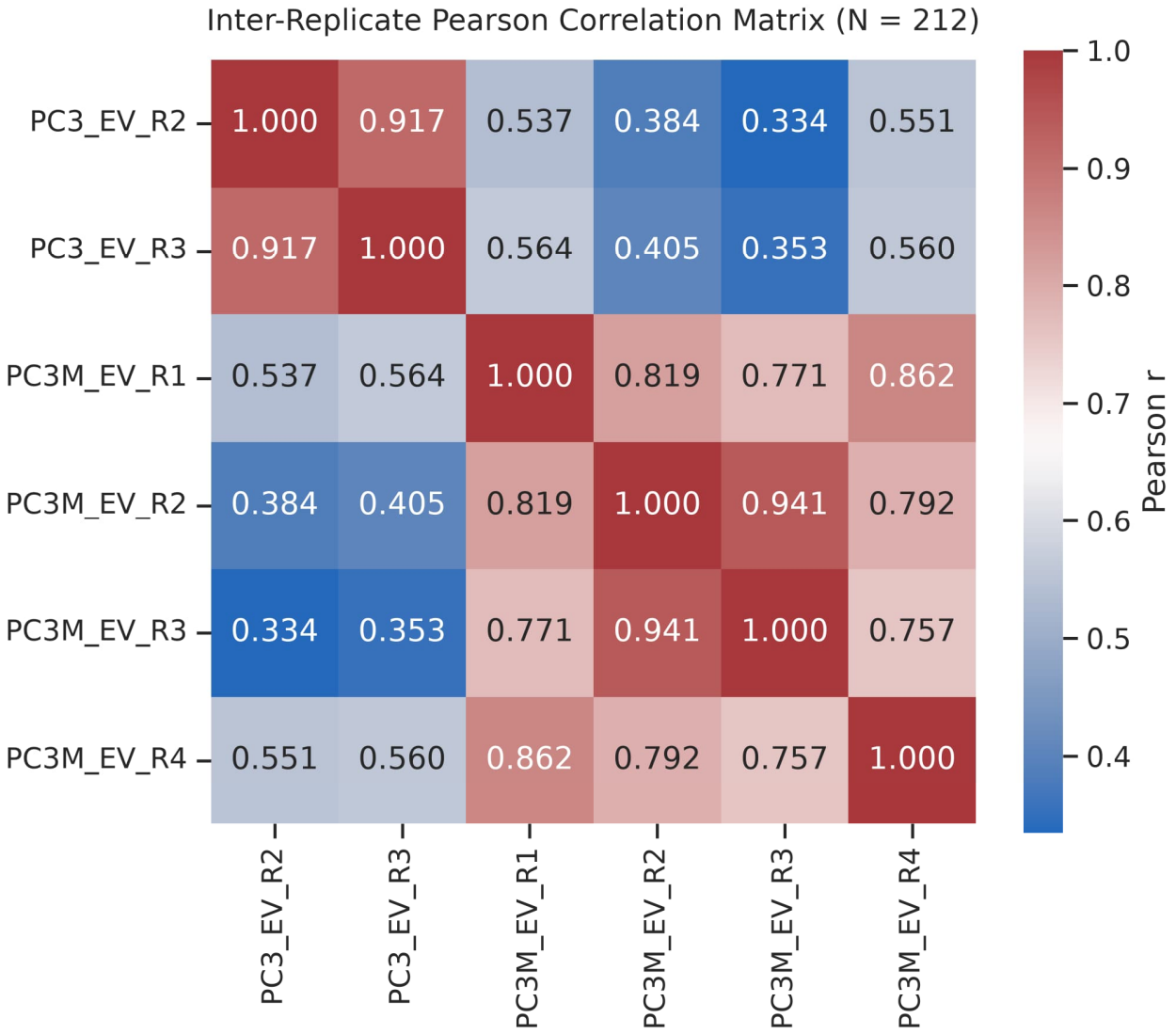


Figure 5. Inter-replicate Pearson cross-correlation matrix of normalized cargo uptake profiles. *Heatmap illustrating pairwise Pearson correlation coefficients (r) computed across all authenticated cargo proteins (N = 212) for the 6 validated biological channels. High intra-condition consistency is observed within both parental PC3 (r = 0.917) and metastatic PC3M (r = 0.76 - 0.94) conditions, accompanied by robust separation between the two phenotypic groups (r = 0.33 - 0.57).*

In the initial exploratory PCA, the first principal component (PC1) captured the overwhelming majority of total dataset variance (87.4%), driving a complete horizontal separation between recipient osteoblasts co-cultured with parental PC3 versus metastatic PC3M vesicles (Figure 6). The second principal component (PC2) accounted for only 5.8% of the total variance, reflecting minor intra-group biological dispersion among PC3M replicates (Figure 6).

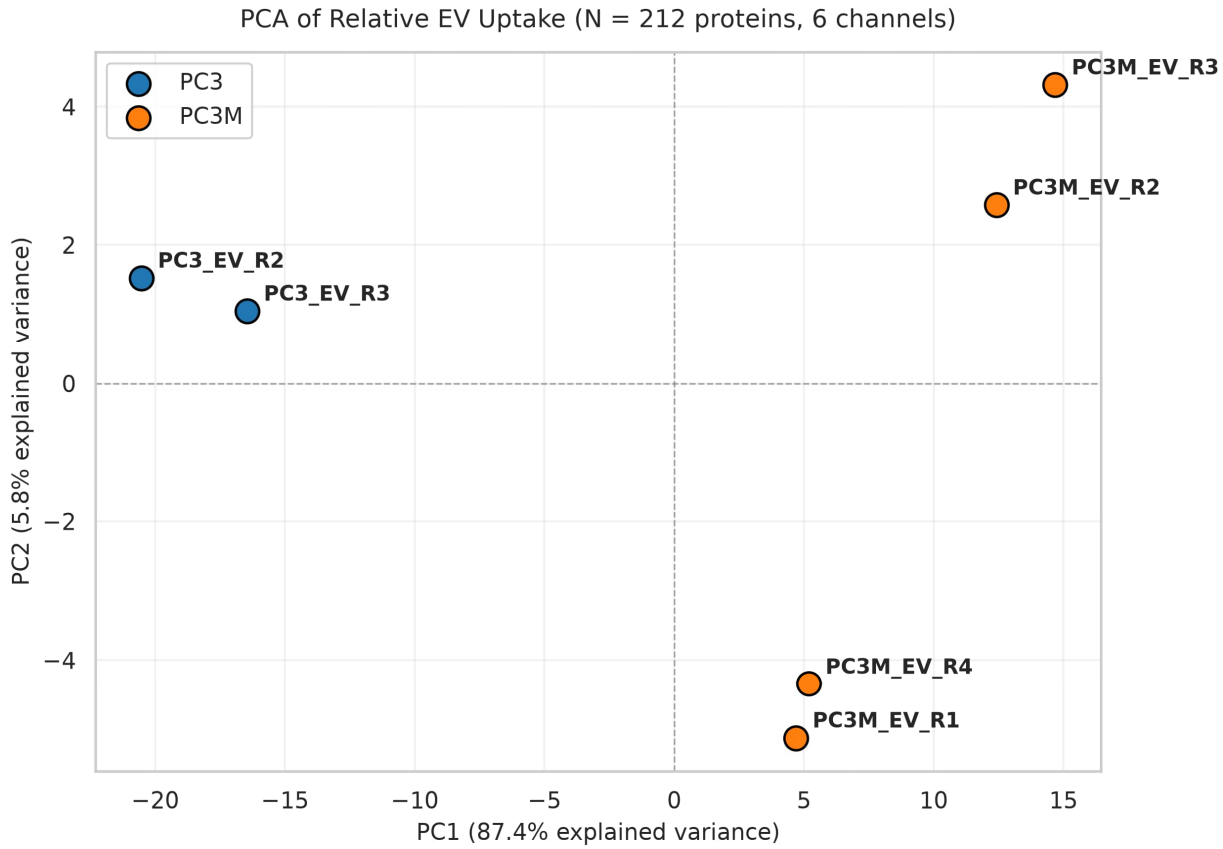


Figure 6. Principal component analysis of relative EV cargo uptake across validated biological replicates. *Score plot of unsupervised PCA conducted on the N = 212 spike-normalized, log_2-transformed EV cargo proteins across 6 experimental channels. PC1 dominates 87.4% of total dataset variance, establishing pronounced baseline separation between recipient osteoblasts (SAOS-2) incubated with parental PC3 EVs (n = 2, blue) and metastatic PC3M EVs (n = 4, orange). PC2 captures 5.8% of variance, representing secondary biological replicate dispersion.*

Identification of the Global Scale Effect

Inspection of the uncentered PC1 feature eigenvectors (loadings) revealed an extreme directional bias across the entire cargo proteome: 211 out of 212 evaluated proteins (99.5%) exhibited strictly positive loading coefficients (ranging up to +0.0734), with only a single protein projecting negatively (-0.0628).

In multivariate linear decomposition, an axis capturing the vast majority of dataset variance where loadings are near-unanimously unidirectional constitutes a canonical Global Scale Effect. Rather than reflecting reciprocal pathway reprogramming or stoichiometric substitution of specific vesicular cargo, PC1 primarily captured a macro-scale phenotypic phenomenon: recipient SAOS-2 osteoblasts internalize metastatic PC3M-derived vesicles at a systematically higher absolute rate and volume than parental PC3vesicles.

While this global scale effect underscores an intrinsically elevated endocytic uptake capacity for metastatic vesicles, this uniform elevation obscures the detection of authentic stoichiometric selectivity. Measured absolute cargo uptake represents the product of cumulative internalized vesicle volume and individual intra-vesicular protein concentrations. Consequently, non-specific bulk uptake artificially inflates reporter signals across virtually all cargo proteins in the metastatic condition.

To distinguish genuine molecular drivers of phenotypic tropism from passive co-internalized passenger cargo, this non-specific bulk volume component must be decoupled prior to downstream statistical hypothesis testing. Furthermore,

because highly internalized structural cargo (such as extracellular matrix components) can exhibit massive asymmetric fold-changes, standard mean-centering would inappropriately distort the proteome-wide baseline. Decoupling must therefore be achieved via Sample-Median Centering, providing mathematical robustness against asymmetric cargo outliers and realigning the analytical baseline to relative stoichiometric composition.

6. Sample-Median Centering and Resolution of Relative Cargo Selectivity

To eliminate the confounding influence of non-specific bulk endocytosis and resolve selective cargo internalization, sample-median centering was applied across all 6 authenticated biological channels. For each channel j, the channel-wide median across all N = 212 quantified proteins was subtracted from each individual protein's $\log_2$-transformed normalized uptake value:

$$y_{ij}^{\text{centered}} = y_{ij} - \text{median}_j(y_j)$$

Operating in logarithmic space makes this transformation mathematically equivalent to dividing raw abundance ratios by the sample median, aligning the central tendency of every channel to an identical baseline of 0.0 on the $\log_2$-scale. This operation cancels out uniform sample-level loading variations, technical channel-to-channel fluctuations, and non-specific bulk internalization volume offsets. Consequently, downstream comparative metrics capture genuine shifts in cargo stoichiometry rather than bulk vesicle consumption.

Following sample-median centering, unsupervised PCA demonstrated a marked redistribution of explained variance across orthogonal dimensions. While the uncentered ordination was heavily dominated by a single scale component (87.4% along PC1), the centered projection rebalanced the variance distribution. The primary axis (PC1) accounts for 50.5% of total dataset variance, maintaining robust segregation between parental PC3replicates (negative coordinates) and metastatic PC3M replicates (positive coordinates). The secondary axis (PC2) explains 20.1% of the variance, resolving underlying biological structure within the PC3M cohort into two distinct biological pairs of replicates (R1/R4 and R2/R3) (Figure 7).

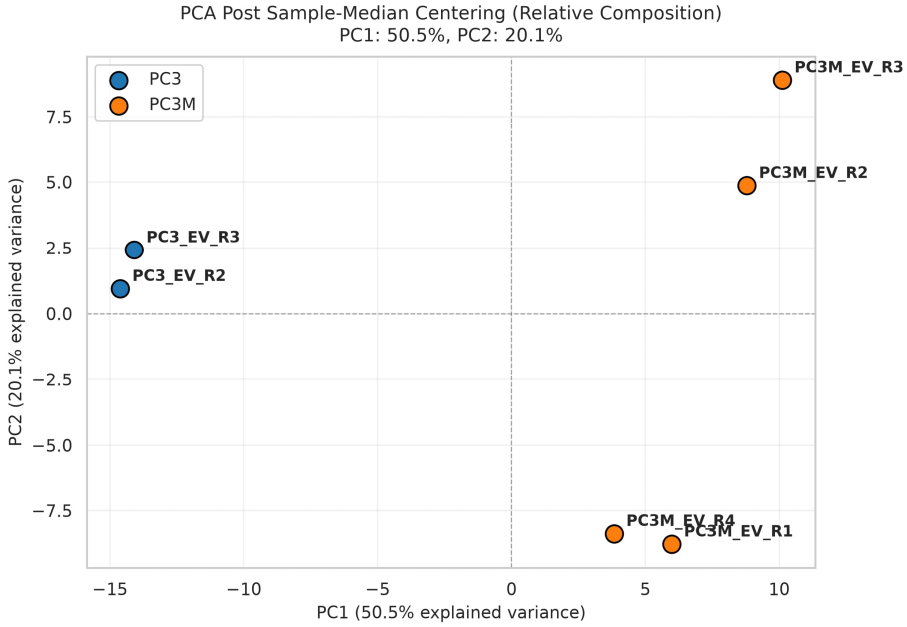


Figure 7. Principal component analysis of relative EV cargo composition following sample-median centering. *Score plot of unsupervised PCA performed on the N = 212 cargo proteins after sample-median centering across the 6 validated biological channels. Subtracting channel medians removes the global scale effect of bulk uptake, reducing PC1 explained variance from 87.4% to 50.5% while preserving robust phenotypic separation between parental PC3(n = 2, blue) and metastatic PC3M (n = 4, orange) conditions. PC2 captures 20.1% of the variance, delineating intra-group biological structure among metastatic replicates.*

In contrast to the uncentered PCA (where 99.5% of proteins displayed positive loadings along PC1) the centered PCA yielded a balanced, bidirectional distribution of feature loadings. Across the quantified proteome, 99 proteins exhibited positive loadings along PC1 (enriched in metastatic PC3M uptake), while 113 proteins projected with negative loadings

(Table 4). This structural transition confirms that sample-median centering successfully decoupled non-specific internalization volume from genuine stoichiometric differences in cargo composition.

Table 4. Extreme Feature Loadings Along PC1 Following Sample-Median Centering (N = 212). Delineates the bidirectional distribution of cargo proteins after uncoupling global bulk scale effects.

Gene Symbol	PC1 Loading	Log_2(Fold Change)
SLC12A2	0.097	0.83
CTNND1	0.096	1.97
COL4A2	0.096	6.01
ANPEP	0.096	1.79
TGM2	0.095	4.13
FN1	0.095	3.76
CCN1	0.094	1.71
HLA-A	0.094	1.03
…		
RPS16	-0.093	-0.87
RPL4	-0.094	-0.82
ATP1A1	-0.095	-0.33
HM13	-0.095	-1.77
TMED9	-0.096	-0.78
PDIA3	-0.096	-0.91
HSPA5	-0.096	-1.04
MAP3K12	-0.096	-2.25

7. Baseline Differential Uptake Screening via Unmoderated Welch's t-Test

Differential uptake between parental PC3(n_1 = 2) and metastatic PC3M (n_2 = 4) conditions was initially evaluated using an unmoderated Welch's two-sample t-test across all N = 212 centered cargo proteins. Unlike Student's formulation, Welch's test accommodates unequal sample allocations and heteroscedastic variances by computing effective degrees of freedom via the Welch–Satterthwaite approximation:

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

Multiplicity adjustment was conducted using the Benjamini-Hochberg False Discovery Rate (FDR) procedure, enforcing discovery thresholds at FDR < 0.10 ($q < 0.10$) and $|\log_2 FC| \geq 1.0$.

This unmoderated baseline screening yielded 22 differentially internalized cargo proteins satisfying both criteria (14 enriched in PC3M uptake and 8 enriched in PC3 uptake) (Figure 8). Cargo displaying significantly elevated uptake in the metastatic condition was dominated by extracellular matrix components (COL4A2, FN1, TNC), cellular adhesion regulators (CTNND1, CCN1), and surface integrin heterodimers (ITGA1, ITGA3, ITGB1). Conversely, cargo enriched under parental PC3 uptake comprised endoplasmic reticulum chaperones and trafficking machinery (HSPA5, PDIA6, LMAN2, HM13), the kinase MAP3K12, and stress-response factors (GDF15).

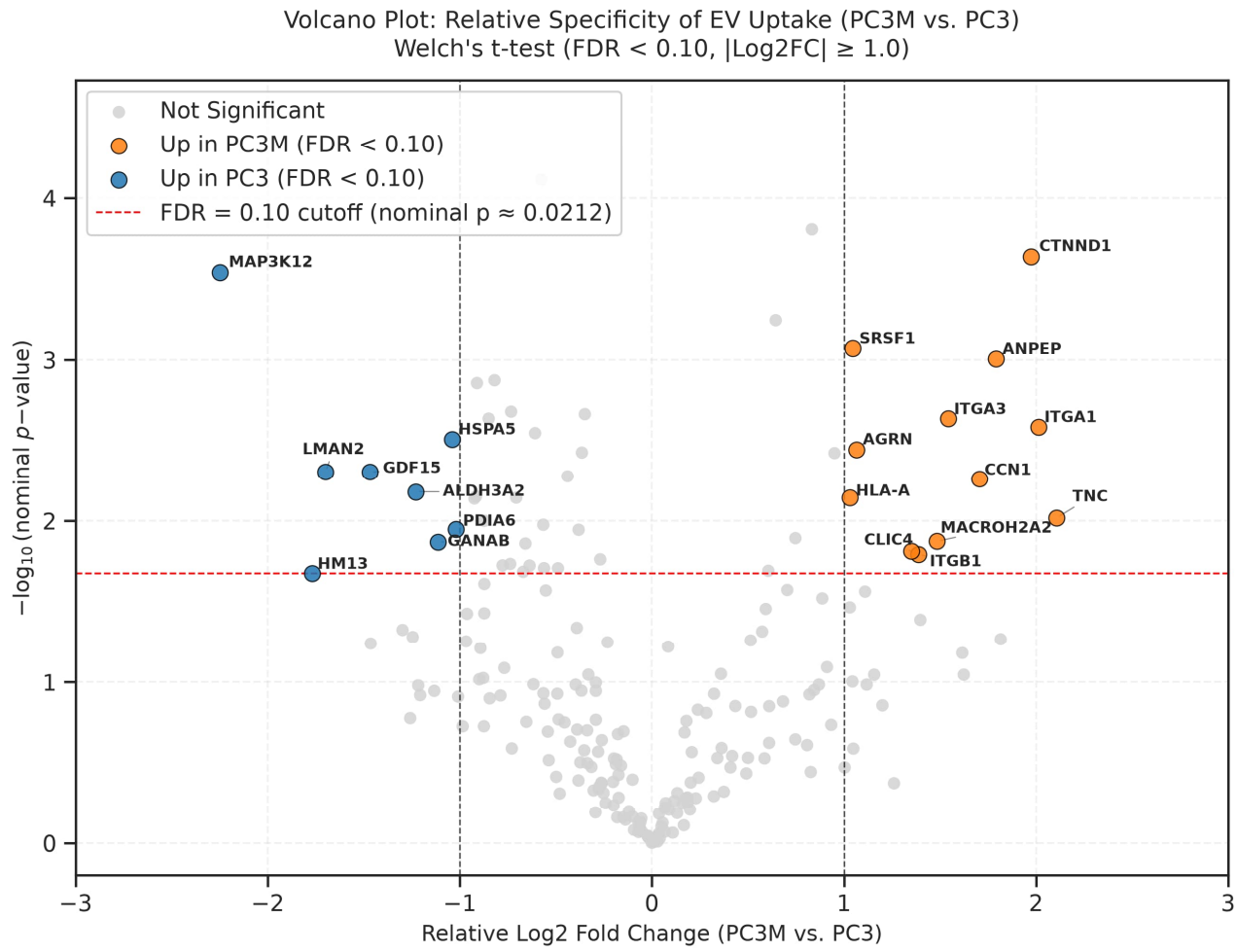


Figure 8. Baseline differential EV cargo uptake screening via unmoderated Welch's t-test. *Volcano plot of relative cargo uptake differences between metastatic PC3M (n = 4) and parental PC3(n = 2) conditions across N = 212 proteins. The horizontal axis indicates relative effect size ($\Delta log_2(FC)$, PC3M vs. PC3), and the vertical axis denotes nominal significance ($-log_{10}$(p-value)). Vertical dashed lines demarcate fold-change thresholds ($|log_2(FC)| \geq 1.0$), while the horizontal red dashed line defines the FDR < 0.10 boundary (nominal $p \approx 0.0212$). A total of 22 proteins pass dual filtering (orange: enriched in PC3M, n = 14; blue: enriched in PC3, n = 8; grey: non-significant, n = 190).*

Analytical Limitations in Constrained Sample Regimes

While Welch's test accurately detects massive-effect outliers, its unmoderated variance denominator exhibits severe instability under limited sample sizes. For the parental PC3 condition ($n_1 = 2$), the residual degrees of freedom collapse to $n_1 - 1 = 1$, restricting total effective degrees of freedom ν to an unstable range ($1 \leq \nu \leq 3$).

In this low-degree-of-freedom regime, sample variance estimates (s_g^2) fluctuate erratically. Peptides exhibiting stochastic, near-identical agreement between the two PC3 replicates yield severely deflated variance denominators, producing artificially inflated t-statistics and false-positive flags. Conversely, true biological drivers exhibiting biologically real effect sizes but modest sample variance are severely penalized under multiplicity correction. A prime example is the matrix cross-linker TGM2 (log_2(FC)= +4.13), which fails to achieve statistical significance under unmoderated testing purely due to small-sample variance instability.

These constraints demonstrate that unmoderated univariate testing lacks the power to reliably resolve EV cargo selectivity in sample-constrained proteomics. This limitation necessitates an Empirical Bayes shrinkage framework that borrows variance information across the global proteome.

8. Empirical Bayes Moderated Differential Uptake Analysis

To overcome the severe loss of statistical power and variance instability inherent to small, unbalanced sample sizes ($n_1 = 2$ vs. $n_2 = 4$, nominal residual degrees of freedom $d_g = 4$), an Empirical Bayes linear modeling framework (limma-trend) was applied across all N = 212 centered cargo proteins.

Individual sample standard deviations (s_g) exhibited pronounced scatter, collapsing toward zero in proteins with stochastic inter-replicate agreement or flaring erratically in high-noise features. Under the Empirical Bayes paradigm, prior technical variance s_{0g}^2 is modeled as a smooth, intensity-dependent function of mean abundance (A_{mean}, on the $\log_2$ scale) using robust regression splines. Posterior residual variances ($\tilde{s}_g^2$) are then calculated by shrinking observed sample variances toward the empirical trend curve:

$$\tilde{s}_g^2 = \frac{d_0 s_{0g}^2 + d_g s_g^2}{d_0 + d_g}$$

This moderation augmented the effective degrees of freedom from $d_g = 4$ to $d_g + d_0 \approx 7.1$, stabilizing the denominator of the moderated t-statistics without introducing arbitrary variance offsets:

$$\tilde{t}_g = \frac{\hat{\beta}_g}{\tilde{s}\sqrt{v_g}}$$

Evaluating statistical significance against the moderated t-distribution followed by Benjamini–Hochberg multiple testing correction (FDR < 0.1, $|\log_2(FC)| \geq 1.0$) expanded the cohort of differentially abundant cargo from 22 (in unmoderated Welch's screening) to 37 high-confidence cargo drivers (Figure 9, Figure 10).

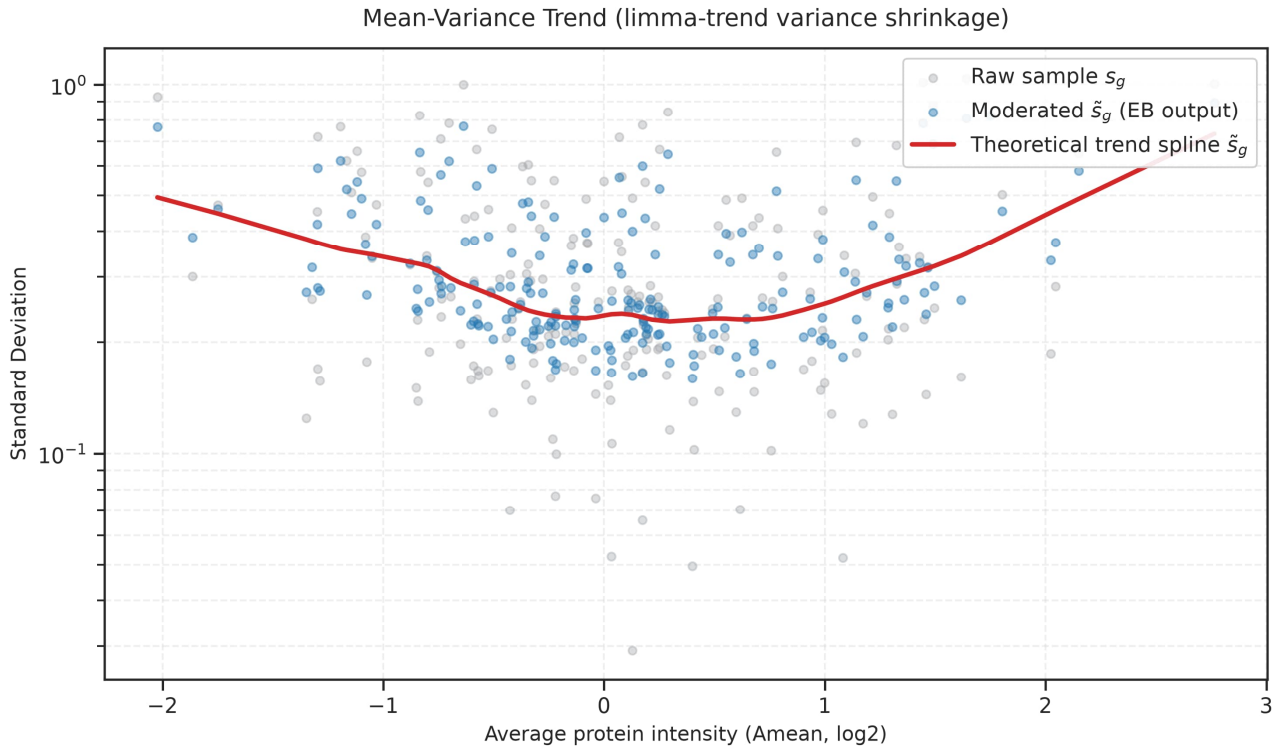


Figure 9. Empirical Bayes variance shrinkage across the EV cargo dynamic range. *Scatter plot displaying the relationship between average protein abundance (Amean, log_2 scale) and residual standard deviation (N = 212). Unmoderated sample standard deviations (s_g, grey circles) exhibit marked scatter due to low degrees of freedom ($d_g = 4$). The solid red curve denotes the intensity-dependent prior variance trend ($s_{0g}(A)$) fitted via cubic regression splines. Moderated standard deviations ($\tilde{s}_g$, blue circles) demonstrate continuous shrinkage toward the empirical prior curve, regularizing the variance denominator.*

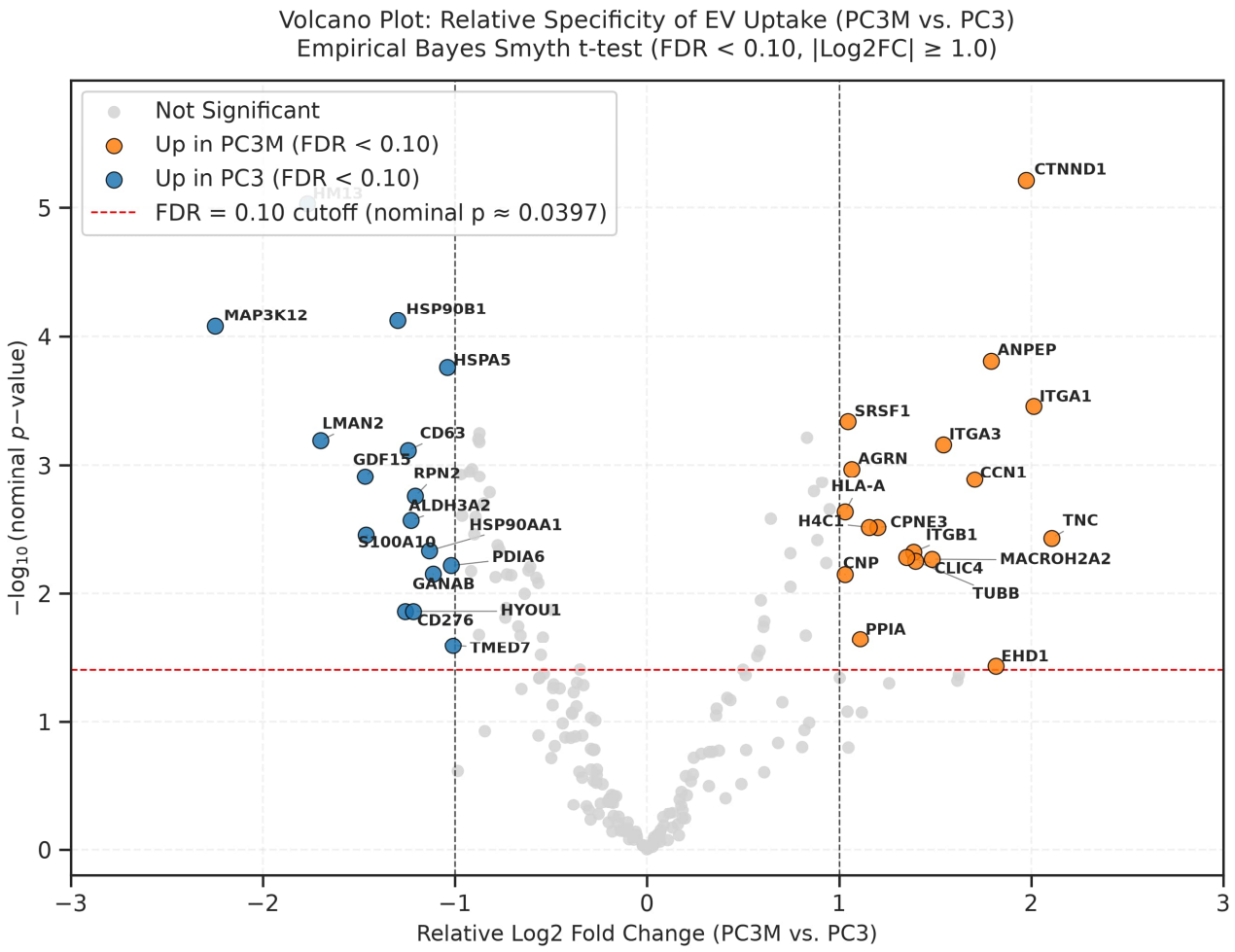


Figure 10. Volcano plot of relative EV cargo uptake evaluated by Empirical Bayes moderated testing. *Relative cargo uptake differences between metastatic PC3M (n = 4) and parental PC3(n = 2) conditions following variance shrinkage. The horizontal axis indicates relative effect size ($log_2(FC)$, PC3M vs. PC3), and the vertical axis denotes statistical significance ($-\ log_{10}$(nominal p-value derived from moderated t-statistics)). Vertical dashed lines demarcate fold-change boundaries ($|log_2(FC)| \geq 1.0$), and the horizontal red dashed line marks the FDR < 0.10 threshold (nominal p ≈ 0.0397). A total of 37 proteins pass both criteria: 21 proteins enriched in metastatic PC3M uptake (orange circles) and 16 proteins enriched in parental PC3uptake (blue circles); non-significant proteins are displayed in grey (n = 175).*

Borrowing variance information across the ensemble retained all of the initial 22 candidate markers identified during Welch's screening while recovering 15 additional statistically robust drivers (+68% expansion in discovery yield) that previously failed multiple-testing thresholds due to residual degree-of-freedom penalties (Table 5)

Table 5. Authenticated EV Cargo Signatures Identified via Empirical Bayes Moderation (FDR < 0.10, $|Log_2(FC)| \geq 1.0$)

Total proteins passing FDR < 0.10 and $\|Log_2(FC)\|$ >= 1.0: 37							
Up in PC3				Up in PC3M			
Gene Symbol	$Log_2(FC)$	$p-value$	q-value	Gene Symbol	$Log_2(FC)$	p-value	q-value
HM13	-1.77	0.00001	0.001	CTNND1	1.97	0.00001	0.001
HSP90B1	-1.30	0.00008	0.003	TGM2	4.13	0.00001	0.001
MAP3K12	-2.25	0.00008	0.003	COL4A2	6.01	0.00005	0.003
HSPA5	-1.04	0.00017	0.005	ANPEP	1.79	0.00016	0.005
LMAN2	-1.70	0.00064	0.009	ITGA1	2.01	0.00035	0.008
CD63	-1.25	0.00077	0.009	FN1	3.76	0.00036	0.008
GDF15	-1.47	0.00123	0.011	SRSF1	1.05	0.00046	0.009
RPN2	-1.21	0.00176	0.013	ITGA3	1.54	0.00069	0.009
ALDH3A2	-1.23	0.00272	0.016	AGRN	1.07	0.00107	0.011
S100A10	-1.46	0.00351	0.019	CCN1	1.71	0.00130	0.011
HSP90AA1	-1.13	0.00466	0.022	HLA-A	1.03	0.00233	0.015
PDIA6	-1.02	0.00609	0.025	H4C1	1.16	0.00308	0.017
GANAB	-1.11	0.00708	0.026	CPNE3	1.20	0.00309	0.017
HYOU1	-1.22	0.01376	0.042	TNC	2.11	0.00373	0.019
CD276	-1.26	0.01378	0.042	ITGB1	1.39	0.00479	0.022
TMED7	-1.01	0.02566	0.069	CLIC4	1.35	0.00526	0.023
				MACROH2A2	1.48	0.00544	0.024
				TUBB	1.40	0.00560	0.024
				CNP	1.03	0.00712	0.026
				PPIA	1.11	0.02276	0.062
				EHD1	1.82	0.03729	0.095

9. Unsupervised Hierarchical Clustering and Phenotype Segregation

Unsupervised hierarchical clustering of the 37 Empirical Bayes-moderated cargo markers across the 6 biological channels yielded a complete primary bifurcation in the column dendrogram, cleanly segregating parental PC3 (PC3) from metastatic (PC3M) EV uptake profiles (Figure 11).

Because clustering was performed agnostic to condition labels, this separation confirms that the derived signature captures authentic phenotypic programs rather than analytical noise or processing artifacts. Within the metastatic arm, the column dendrogram resolved into two paired sub-clades (R2/R3 and R1/R4), directly recapitulating the intra-group biological structure resolved along PC2 in the centered principal component analysis.

The row dendrogram partitioned the 37 proteins into two mutually exclusive, co-regulated modules corresponding to directional uptake tropism (Figure 11):

- **Parental PC3 Module (n = 16):** Displays uniform positive Z-scores across PC3 channels and corresponding suppression across PC3M replicates. This module comprises endoplasmic reticulum lumen chaperones and proteostasis regulators (HSPA5, HSP90B1, HSP90AA1, HYOU1, PDIA6, GANAB, RPN2), stress-response modulators (MAP3K12, GDF15), vesicular trafficking machinery (LMAN2, TMED7, HM13), and the canonical exosomal tetraspanin CD63.
- **Metastatic Module (n = 21):** Displays pronounced relative uptake enrichment across all PC3M replicates. This block encompasses structural extracellular matrix components and cross-linking machinery (COL4A2, FN1, TNC, TGM2), integrin heterodimer subunits (ITGA1, ITGA3, ITGB1), junctional regulators (CTNND1), the matricellular signaling factor CCN1, and cell-surface proteases (ANPEP).

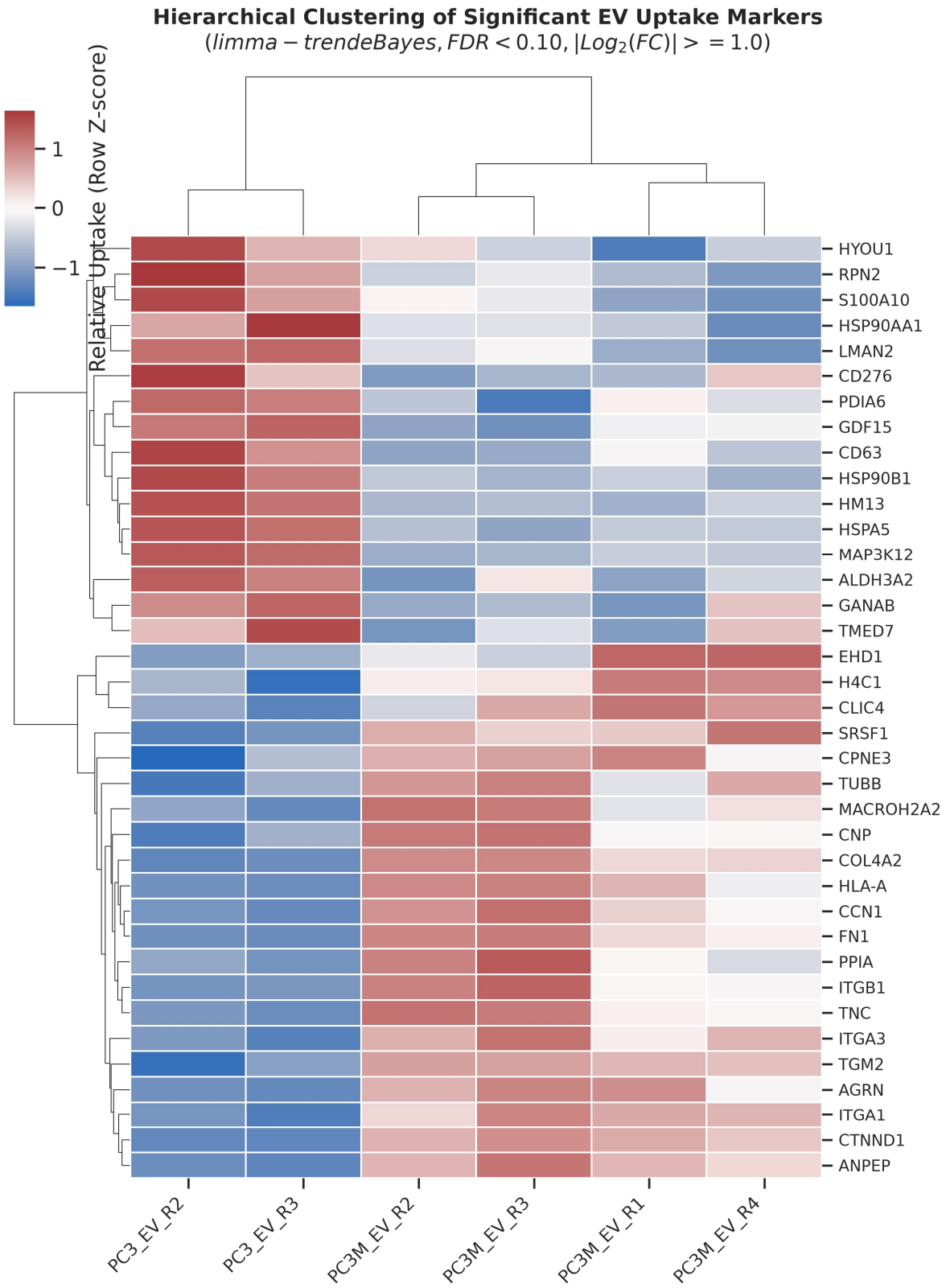


Figure 11. Unsupervised hierarchical clustering of significant EV cargo uptake markers. *Bidirectional heatmap and dendrograms of the 37 differentially internalized proteins identified via Empirical Bayes testing (FDR < 0.10, $|log_2(FC)| \geq 1.0$). Columns correspond to validated biological uptake channels across recipient osteoblasts (SAOS-2) incubated with parental PC3(n = 2) or metastatic PC3M (n = 4) EVs. Rows represent individual cargo proteins, standardized via row-wise Z-scores ($z = (x - \bar{x})/s$; red: relative enrichment, blue: relative depletion). Both dimensions were clustered using Euclidean distance on Z-scores (reflecting correlation distance, $d \propto 1 - r$) and average linkage (UPGMA). The column dendrogram exhibits complete, unsupervised bifurcation between PC3 and PC3M conditions, while the row dendrogram resolves cargo into two distinct programs: a metastatic uptake module (n = 21, bottom) and a parental PC3 uptake module (n = 16, top).*

Surface Drivers vs. Delivered Passengers (Hypothesis)

The unsupervised co-clustering of surface adhesion receptors (integrins $\alpha_1, \alpha_3,$ and β_1) together with bulky structural matrix components (COL4A2, FN1, TNC) points toward a dual-component uptake mechanism:

1. **Active Internalization Drivers:** Membrane-presented integrins and ectoenzymes likely serve as receptor ligands that bind osteoblast-surface counter-receptors, actively docking the vesicle and triggering endocytic internalization pathways.
2. **Co-Internalized Passenger Cargo:** Luminal and matrix-associated cargo (such as collagens and fibronectin) are packaged within or adsorbed onto the vesicle shell, serving as passively co-internalized bioactive cargo that conditions the bone microenvironment post-uptake.

Decoupling active endocytic drivers from passive passenger cargo (through surface-proteome profiling and targeted antibody neutralization assays) constitutes the immediate experimental follow-up.

10. Discussion and Perspectives

Biological Findings and Project Answers

This study addressed whether extracellular vesicle uptake by bone-niche cells is governed by stochastic engulfment or selective cargo internalization. Our data demonstrate that both processes co-exist:

- SAOS-2 cells internalize PC3M vesicles at higher overall volume (global scale effect), yet normalizing for bulk endocytosis confirms that cargo uptake is biologically selective rather than purely passive.
- Parental PC3 uptake is enriched in ER proteostasis machinery and canonical vesicle markers, whereas metastatic PC3M vesicles selectively deliver a microenvironment-remodeling program (basement membrane components, matrix cross-linkers, and adhesion integrins).
- Co-enrichment of adhesion receptors (integrins $\alpha_1, \alpha_3, \beta_1$) with structural ECM proteins (COL4A2, FN1, TNC) suggests a dual mechanism: surface receptors mediate active docking to osteoblasts, while co-internalized matrix elements act as bioactive cargo to remodel the bone niche. Resolving vesicle surface topology from luminal cargo remains a key next step.

Methodological Insights

By anchoring relative uptake to condition-matched internal spike-in standards and implementing sample-median centering, the workflow uncoupled true stoichiometric selectivity from bulk particle loading uncertainty. Downstream ordination topologies and statistical rankings remain mathematically invariant to heuristic particle dosing offsets. Furthermore, borrowing variance information across the proteome via Empirical Bayes moderation stabilized statistical testing in a sample-constrained regime, recovering true effect drivers without inflating false discoveries.

Next Steps:

1. **Isoform Curation:** Group ambiguous multi-mapping peptides at the gene locus level to rescue discarded proteins without introducing assignment bias.
2. **Targeted Pathway Analysis:** Perform functional enrichment strictly against the experimentally detected EV proteome background to prevent whole-genome selection bias.
3. **Rank-Shift Profiling:** Map relative abundance shifts from spike-in baseline to cellular uptake channels to pinpoint low-abundance cargo displaying disproportionately high uptake efficiency.
4. **“Surfaceome” Deconvolution:** Integrate external surfaceome annotations to structurally distinguish functional surface triggers from passively delivered internal cargo.
5. **Functional Validation:** Block candidate surface integrins and ectoenzymes with neutralizing antibodies or siRNA knockdowns to quantify direct inhibition of EV uptake in osteoblasts.